(i) Show that this is a complex.

Now assume that $A = k$ is a field.

(ii) Show that if C or D is an exact sequence, then so is $C \otimes D$.

Hint: Use the decomposition of Exercise 7.8.4.

(iii) Show that any complex C can be identified with a direct sum of an exact sequence and the complex consisting of $H^i(C)$ with the zero differentials, in such a way that the induced isomorphism $H^i(C) \to H^i(C)$ is the identity.

(iv) Show that there is a natural isomorphism of vector spaces

$$H^i(C \otimes D) \cong \bigoplus_{j+m=i} H^j(C) \otimes H^m(D).$$

This is the **Künneth** formula.

7.9. Exact functors

Definition 7.9.1. A functor F between two abelian categories is **additive** if it induces homomorphisms on Hom groups. Also, for k-linear categories one says that F is **k-linear** if it induces k-linear maps between Hom spaces.

It is easy to show that if F is an additive functor, then $F(X \oplus Y)$ is canonically isomorphic to $F(X) \oplus F(Y)$.

Example 7.9.2. The functors Ind_K^G, Res_K^G, $\mathrm{Hom}_G(V, ?)$ in the theory of group representations over a field k are additive and k-linear.

Definition 7.9.3. An additive functor $F : \mathcal{C} \to \mathcal{D}$ between abelian categories is **left exact** if for any exact sequence

$$0 \to X \to Y \to Z,$$

the sequence

$$0 \to F(X) \to F(Y) \to F(Z)$$

is exact. F is **right exact** if for any exact sequence

$$X \to Y \to Z \to 0,$$

the sequence

$$F(X) \to F(Y) \to F(Z) \to 0$$

is exact. F is **exact** if it is both left and right exact.